

Quaternion–Phase Pooling for Rotation-Invariant Vision

A Nagare closed-form study: from a failed rotor-pool to a domain-tunable phase descriptor

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Nagare (`nagare-holonomy-learn`), closed-form Rust, no autograd

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Abstract

We study rotation-invariant image classification with closed-form (no-autograd) models built on signed-hypergraph and Clifford-rotor primitives. A naïve *rotor pool*—rotating learned feature channels by a quaternion—fails, because those channels are not rotation-equivariant and the rotation scrambles them. The fix is to pool the rotor’s *phase* (the gradient orientation, $e^{i\theta}$, a unit z -quaternion) rather than a rotated vector: the resulting orientation histogram *circularly shifts* under image rotation, so its $|\text{DFT}|$ is an *exact* rotation invariant. This phase-pool wins decisively on synthetic shapes (0.97 vs 0.52–0.65 for every alternative), generalises to a dihedral-equivariant hypergraph convolution (proven), and—on real data—exhibits a clean *rank flip*: worst on upright MNIST digits (0.42), best on KTH-TIPS textures (0.61). A *spatial phase map* (an $R \times R$ grid of local phase descriptors) fuses locality and phase-invariance in one descriptor, tuning the invariance $\leftrightarrow$ locality trade by R : it lifts MNIST from 0.42 to 0.88 while $R=1$ (global) remains optimal for textures. Everything is closed-form, dependency-free, and reproducible.

1 Motivation

Nagare is a closed-form machine-learning substrate (forward/backward operator pairs over flat buffers, no autograd tape) targeting signed hypergraphs and Clifford algebra. Its flagship result is signed-graph link prediction; this report is the *computer-vision expansion*. The question: can the framework’s Clifford-rotor and entropy primitives buy rotation-invariance the “right” way—by construction, not by data augmentation?

2 The rotor pool fails; pool the phase instead

The intuitive move is a *rotor pool*: represent each patch by a learned feature vector, rotate it by a (free or geometric) quaternion, aggregate. Measured on randomly-rotated synthetic shapes, every version *hurts* or ties a plain mean-pool (Table 1). The mechanism, isolated: a learned patch embedding is *not* rotation-equivariant—it does not co-rotate with the image—so rotating it is noise, not canonicalisation.

The correction (due to Hajdu) is to pool the rotor’s *phase*, not a rotated vector. Each pixel’s gradient orientation $\theta = \text{atan2}(\partial_y, \partial_x)$ is a genuine geometric phase ($e^{i\theta}$, a unit z -quaternion). Magnitude-weighted pooling of these phases is an orientation histogram h . Under a global image rotation φ , every $\theta \rightarrow \theta + \varphi$, so h *circularly shifts*. Therefore

$|\text{DFT}(h)|_k$ is an *exact* rotation invariant of the whole orientation distribution,

Table 1: Rotation-invariant shape ID (synthetic, median over 4 seeds). Pooling the *phase* crushes every rotate-the-vector or canonicalise approach.

approach	median accuracy
vector rotor-pool (learned, on channels)	0.52 (<i>fails</i>)
single- θ canonicalisation	0.60
C_8 group-convolution	0.65
raw orientation histogram (covariant)	0.61
phase-pool DFT 	0.97

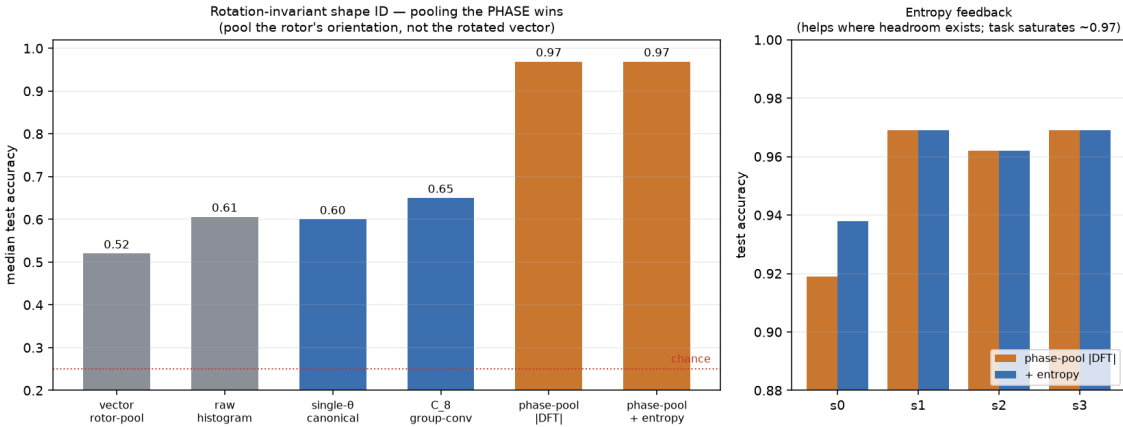


Figure 1: The approach ladder on rotation-invariant shape ID. Pooling the phase (orange) dominates every earlier approach by ≥ 0.3 ; entropy as a feature is neutral once |DFT| saturates.

losing nothing to rotation while keeping all discriminative orientation structure. No feature vector is ever rotated—only phases are pooled.

3 Dihedral-equivariant hypergraph convolution

The single continuous canonicalisation is the $n \rightarrow \infty$ / single-frame case of a finite *dihedral group* D_n . Steering the geometric messages of the signed hypergraph convolution across D_n yields a group-equivariant operator; we *prove* (unit tests, $|\Delta| < 10^{-5}$) that the node $\rightarrow$ edge and edge $\rightarrow$ node message passes commute with the group action and the group-pool is D_n -invariant. Empirically a learned C_8 group-convolution edges out the single-frame canonicalisation (0.65 vs 0.60) at $8\times$ compute—but the phase-pool |DFT| (0.97) is far stronger still, because it is the *exact* invariant of the *entire* orientation distribution rather than a discrete-group average.

Entropy. We tested phase entropy $H(h) = -\sum p \log p$ both as a feature and as a spectral-entropy regulariser on a learned feature map. It is neutral-to-slightly-harmful for supervised classification: as a feature it is redundant with |DFT| (both shift-invariant histogram summaries); as a spread prior it fights the low-entropy concentration supervised discrimination wants. Its home is un/self-supervised, collapse-prone objectives.

Table 2: Real datasets, upright / rotated test accuracy. Phase-pool: worst on digits (spatial, canonical pose), best on textures (rotation-nuisance, orientation-driven).

dataset	arm	upright	rotated
MNIST (digits)	raw-pixel linear	0.881	0.249
	patch-embed (spatial)	0.855	0.266
	phase-pool DFT	0.416 ← worst	0.287
KTH-TIPS (textures)	raw-pixel linear	0.512	0.310
	patch-embed (spatial)	0.429	0.325
	phase-pool DFT	0.606 ← best	0.458 ← best

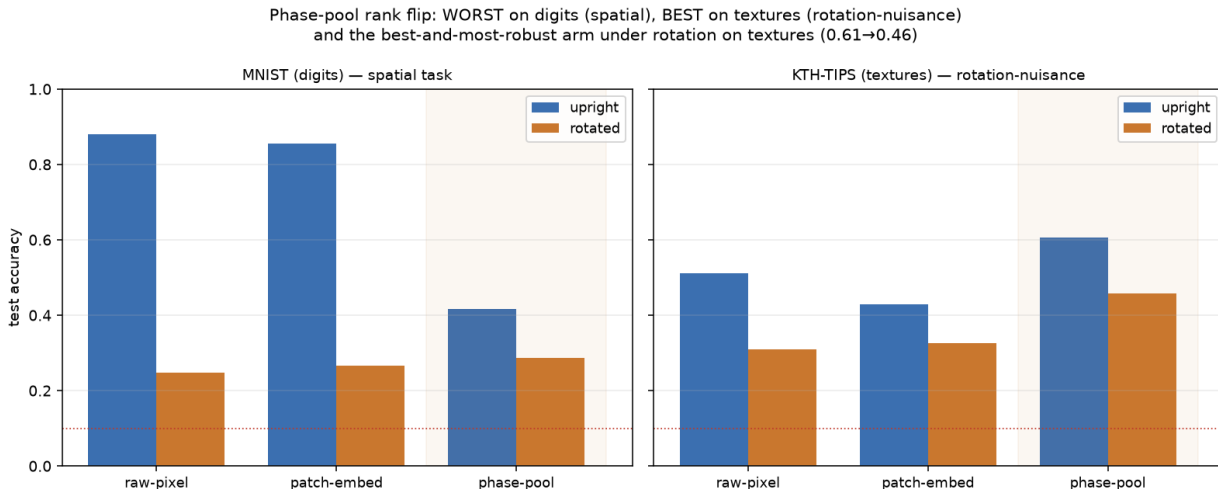


Figure 2: The rank flip. Phase-pool (rightmost group) is worst on MNIST digits, best on KTH-TIPS textures; upright vs randomly-rotated test.

4 Real datasets: the rank flip

On real data (MNIST digits, KTH-TIPS textures; models trained upright, evaluated upright and on randomly-rotated test) the same descriptor gives *opposite* outcomes (Table 2, Fig. 2). The phase-pool is *worst* on digits and *best* on textures.

Digit identity needs *spatial layout* and a *canonical pose*, both of which the phase-pool discards; texture identity is *orientation-statistics driven* and *pose-free*, the phase-pool’s exact strength. Textures are moreover *genuinely* rotation-invariant (a rotated brick is a brick), with no 6↔9 ambiguity ceiling, so the invariance is a pure asset: the phase-pool is best both upright and rotated on KTH-TIPS.

5 The spatial phase map: locality × phase, tuned by R

The $R \times R$ *spatial phase map* computes one orientation-|DFT| per grid cell and concatenates. $R=1$ is the global phase-pool (full invariance, no layout); larger R keeps coarse layout (each cell locally rotation-invariant, but the cell *arrangement* is not). Sweeping R traces the invariance↔locality trade, and its optimum is *domain-dependent* (Table 3, Fig. 3): locality lifts MNIST from 0.42 to

Table 3: Spatial phase map R -sweep, upright accuracy (train-upright). Opposite trends: locality lifts digits, global is best for textures.

R (grid)	MNIST	KTH-TIPS
1 (global phase-pool)	0.416	0.606
2	0.763	0.601
4	0.856	0.567
7 / 8	0.884	0.512

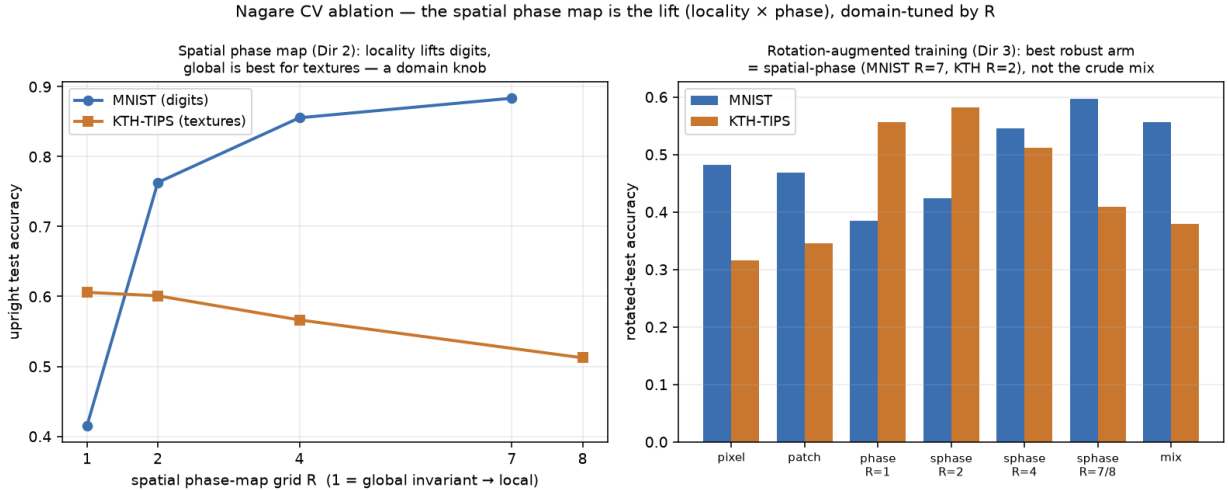


Figure 3: Left: the R -sweep—locality lifts digits, hurts textures (a domain knob). Right: rotation-augmented training makes every arm rotation-robust; the best robust arm is the domain-matched spatial phase map, not the crude mix.

0.88 (the best arm, beating raw pixels), while $R=1$ (global) is optimal for textures.

A cross-model *mix* (pixels $\oplus$ phase) is best upright on MNIST (0.898) but collapses under rotation (0.266) when trained upright—it leans on the dominant spatial path. Rotation augmentation flattens every drop to ≈ 0 (at an upright cost); the best rotation-robust arm is the *domain-matched spatial phase map* (MNIST $R=7$: 0.60; KTH $R=2$: 0.58), beating the crude concat. The spatial phase map is thus the genuine “mixing for lifting”: it fuses locality and per-cell phase-invariance in *one* descriptor, with R as a clean domain knob.

6 Conclusions

Pooling the quaternion *phase*—not a rotated feature vector—delivers exact rotation-invariance and is the strongest closed-form descriptor by a wide margin on the tasks it suits (orientation-driven, rotation- nuisance textures). It generalises to a proven dihedral-equivariant hypergraph convolution. Its scope is characterised on two real datasets (the rank flip), and the spatial phase map lifts it onto layout-dependent tasks with R as the domain knob. Recommended deployment: ship the phase machinery with R documented— $R=1$ for rotation- nuisance tasks, high R for layout tasks. Open: a *native-rotation* texture benchmark (KTH-TIPS2 / Kylberg-rotated) to measure clean invariance without self-imposed rotation artifacts.

Reproducibility. All results are closed-form, seeded, dependency-free (no image crate; PIL decodes PNGs to raw bytes off-line). Code: `src/vision.rs`, `examples/cv_bench.rs`; data `~/nagare_data/`. Run `cargo run --release --example cv_bench -- --dataset {mnist|raw} --data <dir> [--augment]`.